ETHICAL HACKING

PREMIUM COURSE SYLLABUS

Beginner → Advanced



CyberTrix

COURSE ROADMAP

18 modules • practical labs • assessment workflow • professional reporting

MODULE 01 Introduction to Ethical Hacking

MODULE 02 Networking Fundamentals

MODULE 03 Linux Fundamentals for Pentesters

MODULE 04 Reconnaissance & Information Gathering

MODULE 05 Scanning & Enumeration

MODULE 06 Vulnerability Assessment

MODULE 07 Web Application Fundamentals

MODULE 08 Web Application Pentesting

MODULE 09 Password Security & Authentication

MODULE 10 System & Service Security

MODULE 11 Exploitation Fundamentals

MODULE 12 Privilege Escalation

MODULE 13 Active Directory Fundamentals

MODULE 14 Active Directory Pentesting

MODULE 15 Wireless Security Fundamentals

MODULE 16 Network Traffic Analysis

MODULE 17 Post-Exploitation & Security Assessment

MODULE 18 Vulnerability-to-Report Workflow

MODULE 01

Introduction to Ethical Hacking

- What is ethical hacking?
- Types of hackers
- Black box / Grey box / White box
- Legal & ethical boundaries
- Setting up a legal practice lab
- Cybersecurity vs ethical hacking
- Types of penetration testing
- Rules of engagement
- Pentesting methodology

LAB:
Set up a legal, isolated practice environment.

MODULE 02

Networking Fundamentals

- What is a network?
- MAC addresses
- Ports & protocols
- DNS
- HTTP / HTTPS
- LAN / WAN
- IPv4 & IPv6
- TCP vs UDP
- DHCP
- Common networking commands
- IP addresses
- Subnetting basics
- TCP 3-way handshake
- ARP

LAB:
Explore and troubleshoot a controlled lab network.

MODULE 03

Linux Fundamentals for Pentesters

- Linux filesystem
- Users & groups
- Processes
- Package management
- SSH
- Important Kali Linux commands
- Terminal basics
- File permissions
- Services
- Environment variables
- Bash basics
- Installing & managing tools

LAB:
Build and manage a Kali Linux pentesting workspace.

MODULE 04

Reconnaissance & Information Gathering

- Passive vs active reconnaissance
- DNS enumeration
- Subdomain discovery
- Search-engine reconnaissance
- OSINT fundamentals
- Organizing reconnaissance data
- Domain information
- WHOIS concepts
- Technology identification
- Metadata
- Building a reconnaissance workflow

LAB:
Recon a deliberately vulnerable practice target.

MODULE 05

Scanning & Enumeration

- Host discovery
- TCP/UDP scanning
- OS detection
- Nmap fundamentals
- Enumeration methodology
- Port scanning concepts
- Service detection
- Banner grabbing
- Nmap scripting basics
- Interpreting scan results

LAB:
Scan and enumerate an authorized lab network.

MODULE 06

Vulnerability Assessment

- Vulnerability vs exploit
- False positives
- Manual verification
- Reading vulnerability reports
- Documentation
- CVE & CVSS
- Vulnerability scanners
- Prioritizing vulnerabilities
- Finding attack paths

LAB:
Validate findings against intentionally vulnerable targets.

MODULE 07

Web Application Fundamentals

- How websites work
- HTTP request/response
- Headers
- Cookies
- Authentication
- Same-origin concepts
- Client/server architecture
- Methods
- Status codes
- Sessions
- Authorization
- Burp Suite fundamentals

LAB:
Inspect requests and responses in a controlled web application.

MODULE 08

Web Application Pentesting

- Information disclosure
- IDOR
- Command injection concepts
- SSRF concepts
- Business logic vulnerabilities
- Authentication weaknesses
- XSS
- File inclusion
- CSRF
- Authorization issues
- SQL injection
- File upload vulnerabilities
- Security misconfiguration

LAB:
Practice against intentionally vulnerable applications.

MODULE 09

Password Security & Authentication

- Password storage
- Password policies
- Wordlists
- Offline password auditing in a lab
- Rate limiting
- MFA
- Hashing vs encryption
- Authentication mechanisms
- Hash identification
- Brute-force concepts
- Account lockout

LAB:
Audit only provided lab credentials and hashes.

MODULE 10

System & Service Security

- Common services
- FTP
- HTTP
- Service misconfiguration
- Vulnerable software
- Identifying attack paths in a lab
- SSH
- SMB
- Database services
- Weak credentials
- Security hardening concepts

LAB:
Assess services on authorized vulnerable machines.

MODULE 11

Exploitation Fundamentals

- What is exploitation?
- Vulnerability → exploit → access
- Exploit modules
- Sessions
- Understanding exploit output
- Exploit vs payload
- Metasploit fundamentals
- Payload concepts
- Basic post-exploitation concepts
- Safe exploitation inside a lab

LAB:
Validate vulnerabilities only inside controlled environments.

MODULE 12

Privilege Escalation

- LINUX — SUID/SGID
- LINUX — PATH issues
- LINUX — Environment variables
- WINDOWS — Services
- WINDOWS — Registry concepts
- LINUX — File permissions
- LINUX — Scheduled tasks
- LINUX — Kernel vulnerabilities WINDOWS
- — Permissions
- WINDOWS — Common misconfigurations
- LINUX — Sudo misconfigurations
- LINUX — Services
- WINDOWS — Users & groups
- WINDOWS — Scheduled tasks
- WINDOWS — Privilege escalation methodology

LAB:
Escalate privileges on intentionally vulnerable machines.

MODULE 13

Active Directory Fundamentals

- What is Active Directory?
- Domain Controller
- Organizational Units
- LDAP
- Group Policy
- AD enumeration methodology
- Domain / forest / tree
- Users & groups
- Kerberos fundamentals
- NTLM concepts
- Windows authentication

LAB:
Build a small controlled Active Directory practice lab.

MODULE 14

Active Directory Pentesting

- Domain enumeration
- Group enumeration
- Kerberos-related attack concepts
- Misconfigured permissions
- Lateral movement concepts
- User enumeration
- Share enumeration
- Credential exposure
- Privilege escalation paths
- Attack-path analysis

LAB:
Assess a controlled AD environment.

MODULE 15

Wireless Security Fundamentals

- Wi-Fi architecture
- 2.4 GHz / 5 GHz
- WPA2 / WPA3
- Wireless reconnaissance
- Wireless security assessment
- SSID / BSSID
- Authentication
- Encryption
- Monitoring concepts
- Securing wireless networks

LAB:
Assess only your own or explicitly authorized test network.

MODULE 16

Network Traffic Analysis

- Packet structure
- TCP/IP packet analysis
- HTTP traffic
- TCP streams
- PCAP analysis
- Wireshark
- DNS traffic
- ARP traffic
- Identifying suspicious traffic
- Network troubleshooting

LAB:
Analyze supplied PCAP files and controlled lab traffic.

MODULE 17

Post-Exploitation & Security Assessment

- Understanding obtained access
- Identifying sensitive information
- Persistence concepts
- Data exposure assessment
- Evidence collection
- System enumeration
- Privilege assessment
- Lateral movement concepts
- Cleanup

CONTROLLED LABS ONLY:
Focus on authorized environments and assessments.

MODULE 18

Vulnerability-to-Report Workflow

- Finding identification
- Evidence collection
- Business impact
- Remediation
- Executive summary
- Reproduction steps
- Risk rating
- Technical impact
- Screenshots
- Professional pentest report structure

OUTCOME:
Turn technical findings into a professional penetration-testing report.

CYBERTrix OUTCOME

FROM LEARNER → PENTESTER

A complete authorized assessment workflow, from discovery to professional reporting.

01 **Reconnaissance**
Collectandorganizetarget information.

02 **Enumeration**
Maphosts,ports,services and technologies.

03 **Assessment**
Validatevulnerabilities and prioritize risk.

04 **Exploitation**
Safelyvalidatefindings inside controlled labs.

05 **Privilege & Analysis**
Assess impact and attack paths.

06 **Reporting**
Produceevidence, risk and remediation guidance.

CYBERTrix

LEARN • PRACTICE • BUILD • REPORT

AUTHORIZED LABS ONLY